

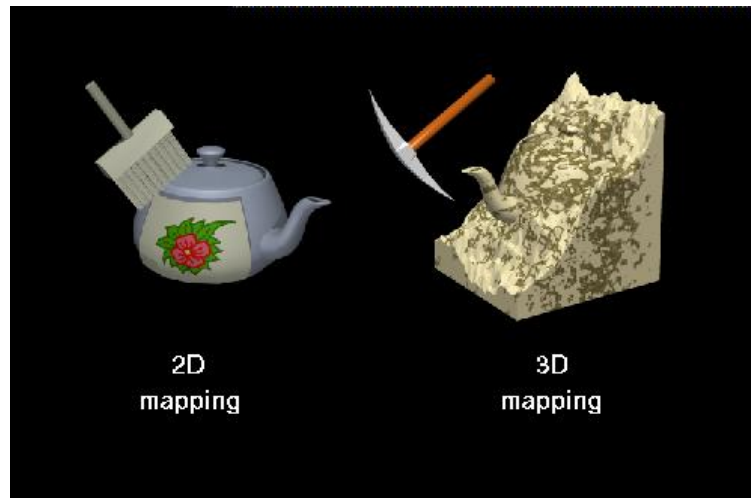
Texture Mapping (2)

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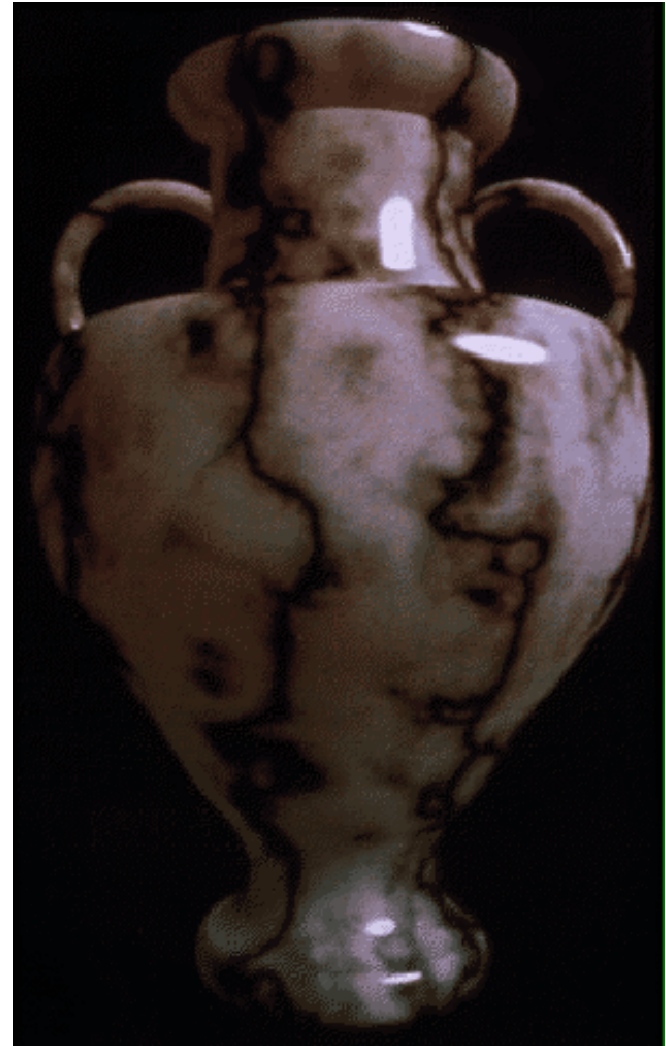
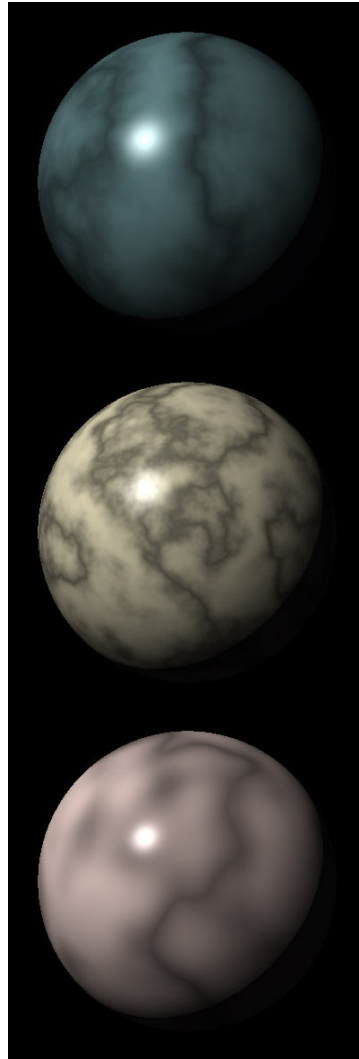
3D texture mapping

- ❑ Developed by D.R. Peachey in 1983 and by K. Perlin in 1985
- ❑ Texture – texture value exists everywhere in the object domain
- ❑ Given a point on the surface of an object (x_w, y_w, z_w) the texture value is given by the identity mapping $T(x_w, y_w, z_w)$
- ❑ Equivalent with sculpting an object out of a block of material
- ❑ The color of the object is determined by the intersection of its surface with the predefined 3D texture field
- ❑ Usually the 3D texture space is obtained by procedural generation



Examples from the course Computer Graphics, by Pascal Vuylsteker, AN Univ., 2003.

3D texture mapping - examples



3D texture mapping – example

Wood grain simulation

1. A cylinder:

$$r_1 = (u^2 + w^2)^{1/2}$$

2. Perturb r with a sinusoidal function

$$r_1 = r_1 + 2 \sin(a\theta)$$

3. Twist along the axis of the cylinder

$$r_1 = r_1 + 2 \sin(a\theta + v/b)$$

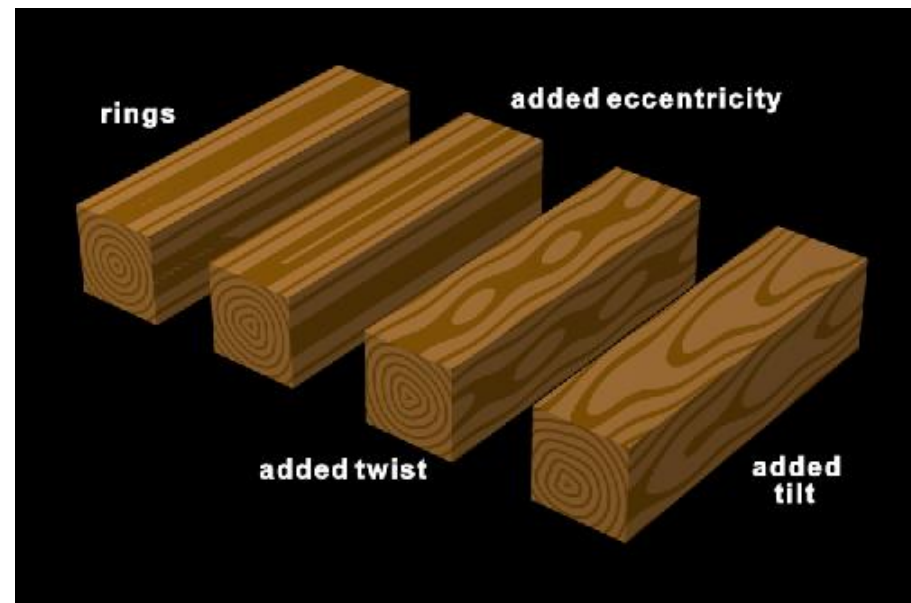
Where a , b are constants

$$\theta = \tan^{-1}(u/w)$$

4. Index the texture

$$(x,y,z) = \text{tilt}(u,v,w)$$

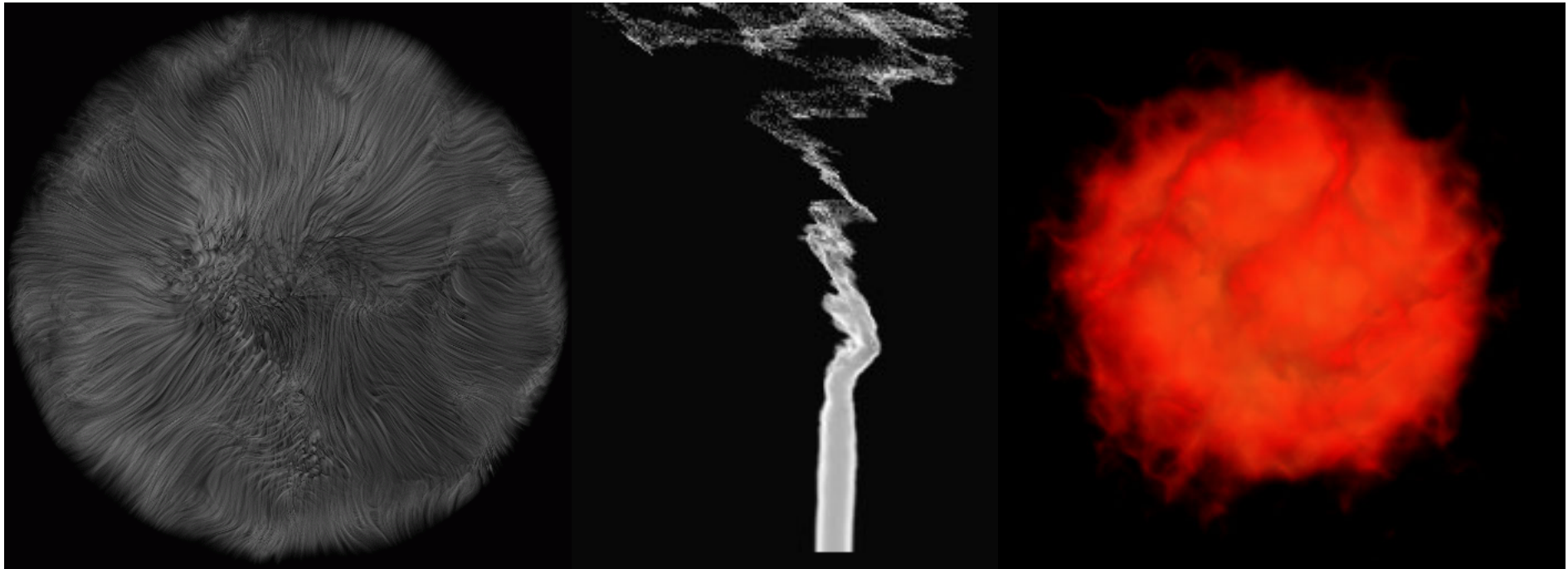
Where “tilt” is a function implementing any rotation between the two coordinate systems



Examples from the course Computer Graphics, by Pascal Vuylstekker, AN Univ., 2003.

Hypertextures

- ❑ Hypertextures – 3D textures introduced by K. Perlin in 1989
- ❑ Objects do not have surface
- ❑ The object surface is defined by a density function and the texture is evaluated within the whole object space
- ❑ Example of modeling: hair, fur, fire, glass, flowing fluids, etc.



Reference: Textures produced by Ken Perlin, NYU Media Research Lab

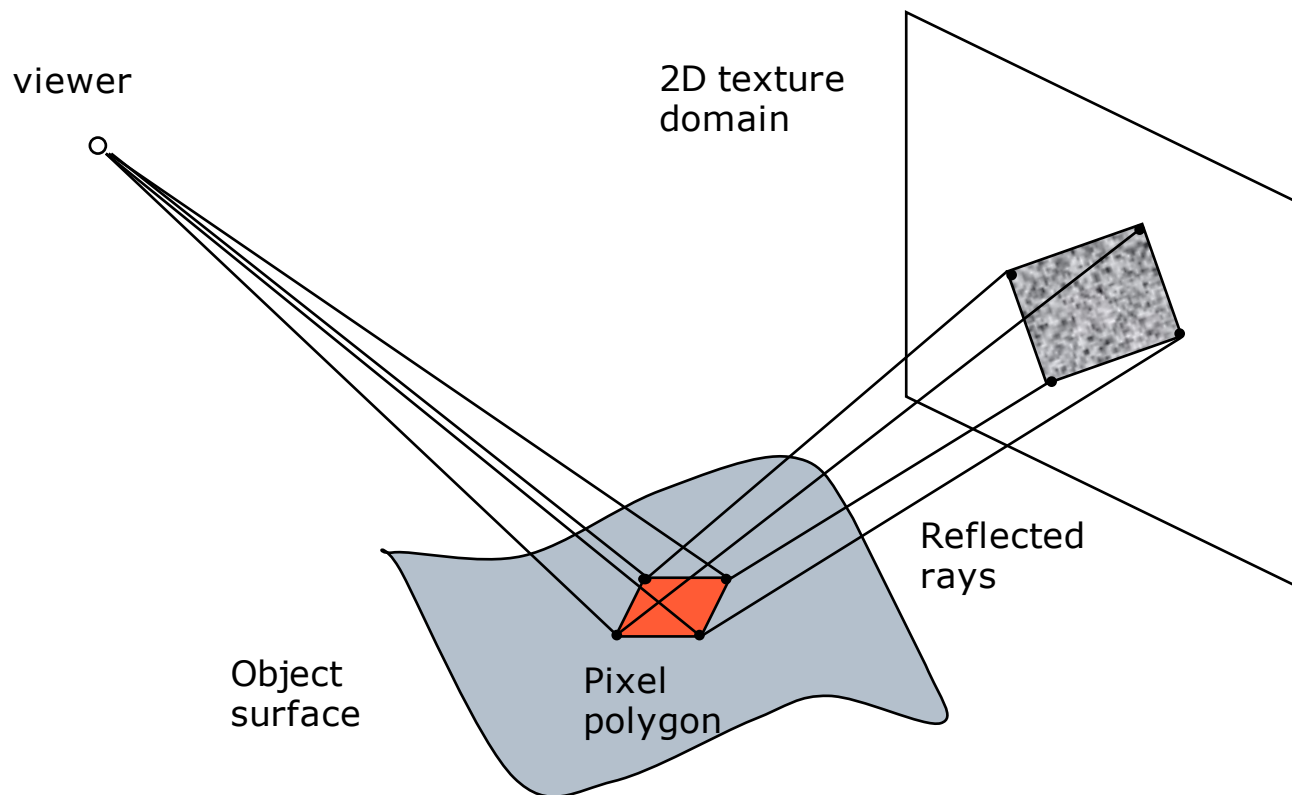
3D texture and noise

Examples from the course Computer Graphics, by Pascal Vuylsteker, AN Univ., 2003.



View dependent texture mapping

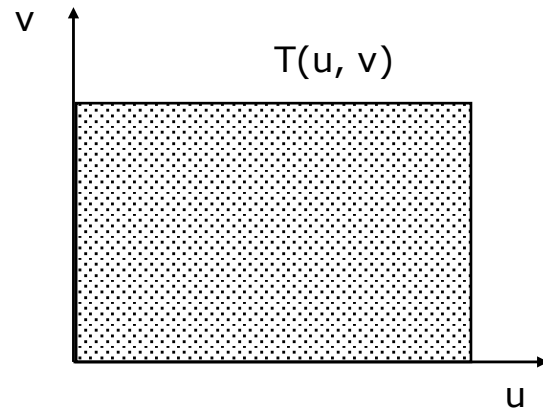
- Texture mapping by reflection (chrome mapping) and refraction of viewing ray
- Ex. Environment mapping



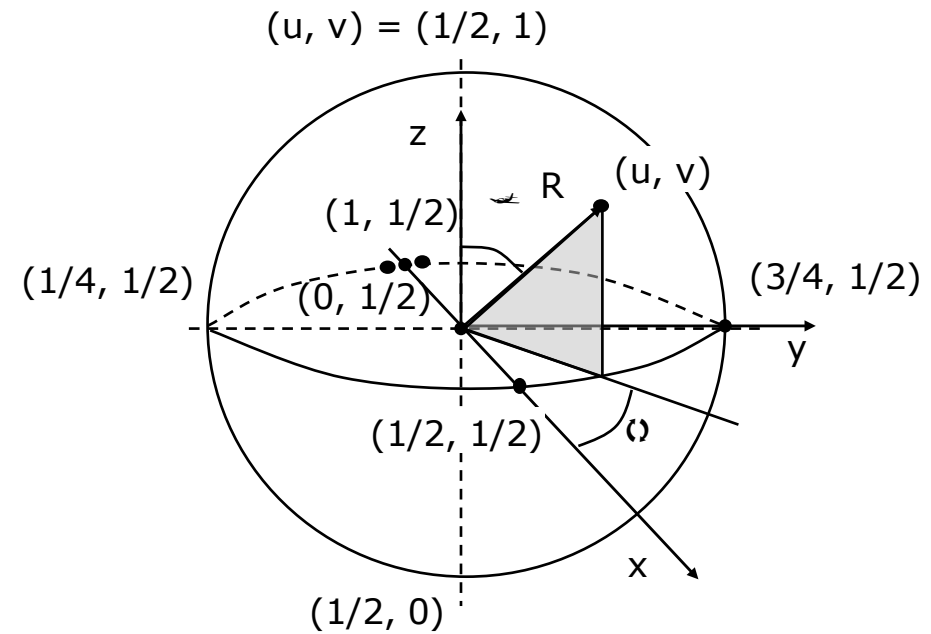
View dependent mapping - examples



Chrome mapping - example

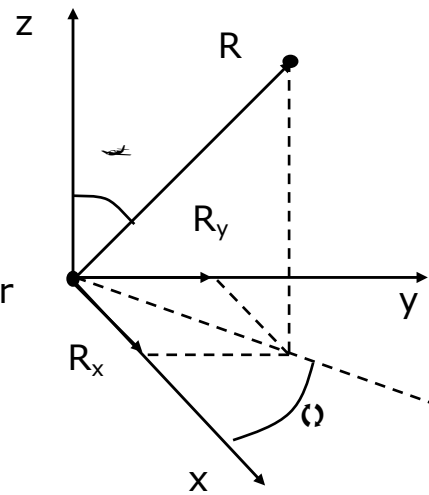


(a) Texture domain



(b) Intermediate surface mapping

(c) Specular reflection vector in spherical coordinates



Chrome mapping - example

Chrome mapping through an intermediate surface as a sphere

Inverse mapping:

$$(u, v) = ((\theta + \pi)/2\pi, (\varphi + \pi/2)/\pi), \quad u, v \in [0, 1]$$

where:

- $\pi < \theta \leq \pi$ azimuth angle
- $-\pi/2 < \varphi \leq \pi/2$ vertical angle

Considering the unitary R vector, it can be written:

- $-\pi < \theta = \arctg(R_y / R_x) \leq \pi$
- $-1 < R_z \leq 1$

Then the inverse mapping based on the reflection vector:

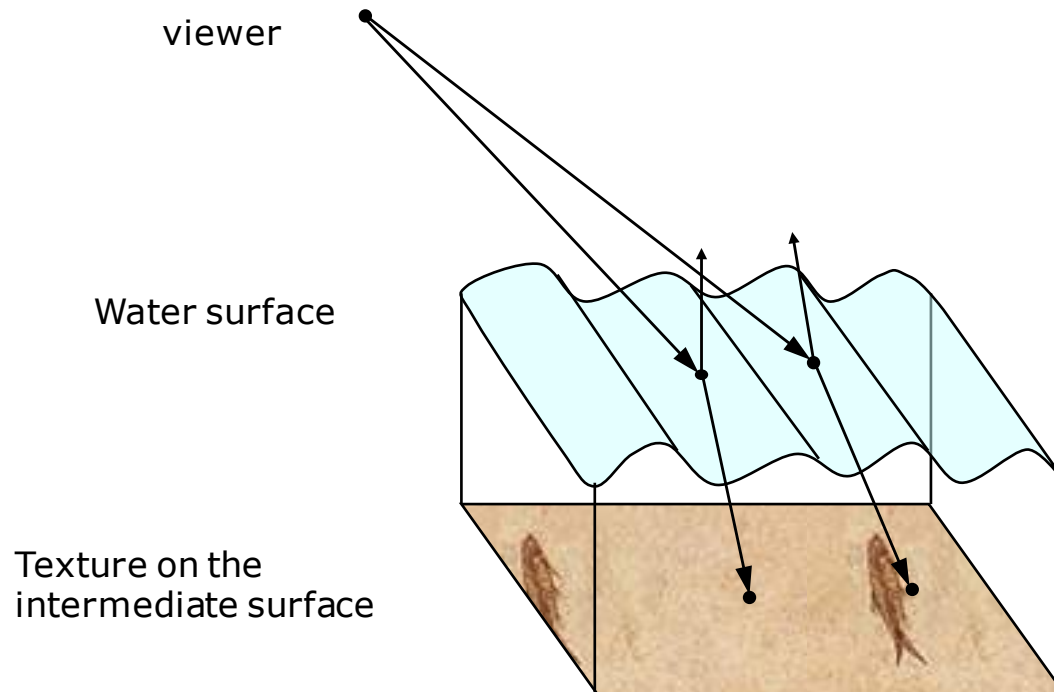
$$u = 1/2 (1 + 1/\pi \arctg(R_y / R_x))$$
$$v = 1/2 (1 + R_z)$$

Chrome mapping approach:

1. Compute the normal vector N for the P point on the object
2. Compute the specular reflection vector R as a function of N and viewer's position
3. Compute the u and v coordinates in the texture domain

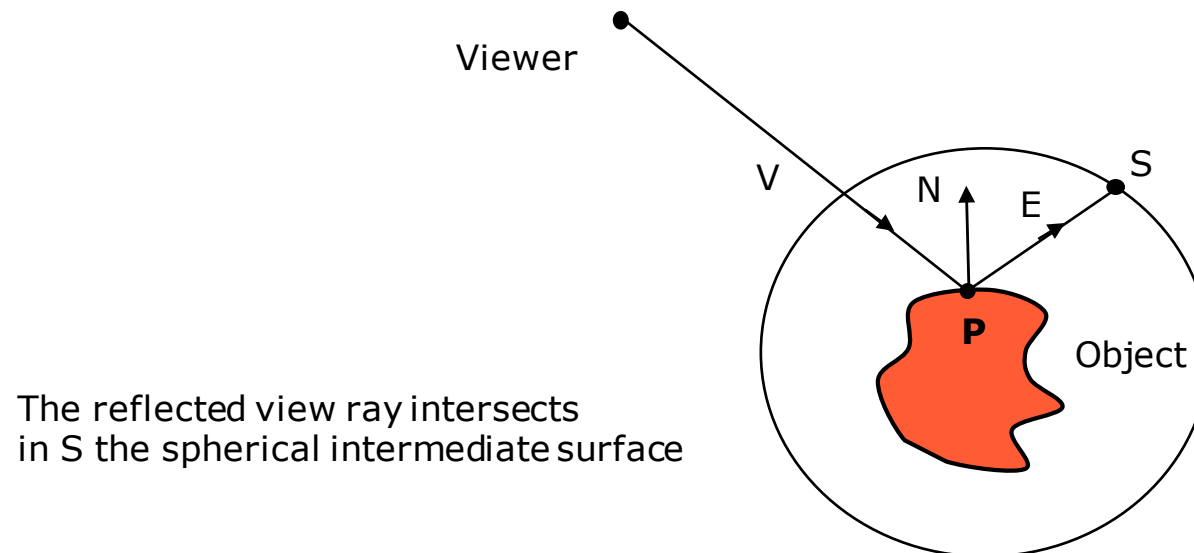
Refraction mapping

- ❑ Compute the refracted rays
- ❑ Intermediate surface is a plane hit by the refracted ray



Environment mapping

- ❑ Introduced by J.F. Blinn and M.E. Newell in 1976. The intermediate surface is the interior surface of a sphere
- ❑ In 1984, G.S. Miller and C.R. Hoffman, and in 1986, N. Greene and P.S. Heckbert used as intermediate surface a cube
- ❑ A simplification of the ray tracing method



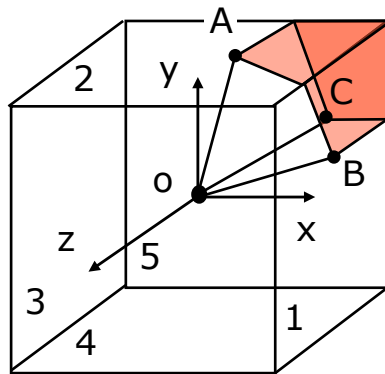
Environment mapping - example



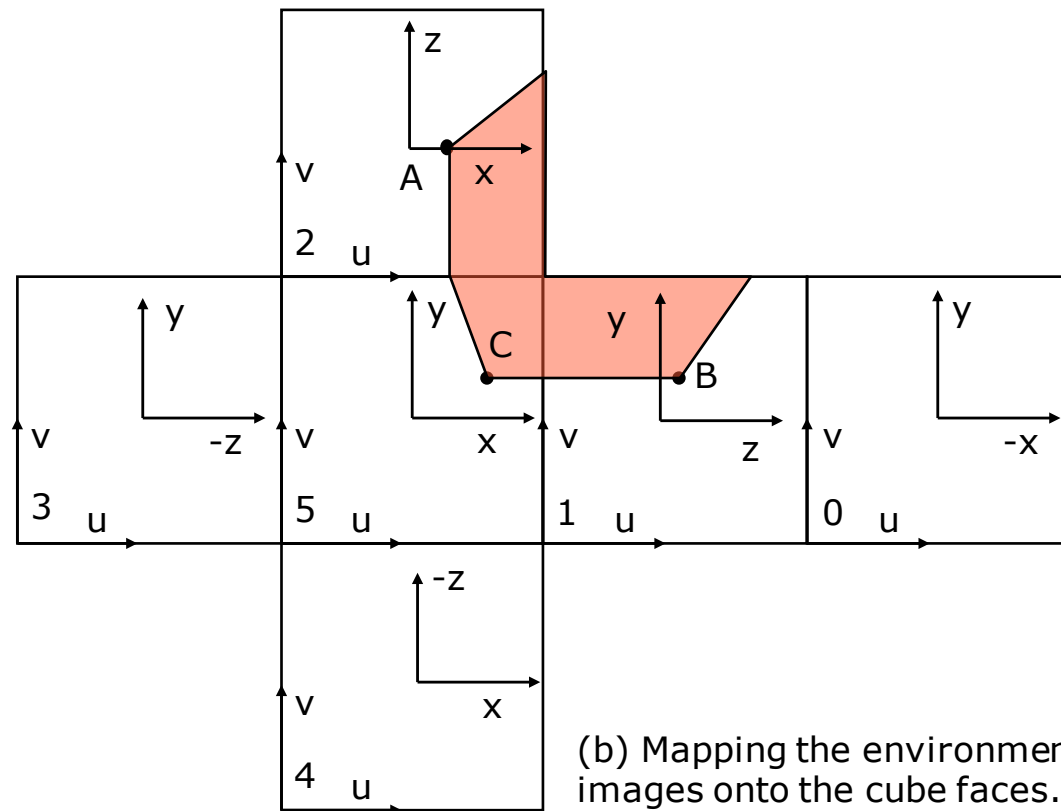
Environment mapping by a cube

- Transformation of the point $(x, -0.5, z)$ on the normalized cube into texture domain uv :

$$u = x + 0.5, v = -z + 0.5.$$

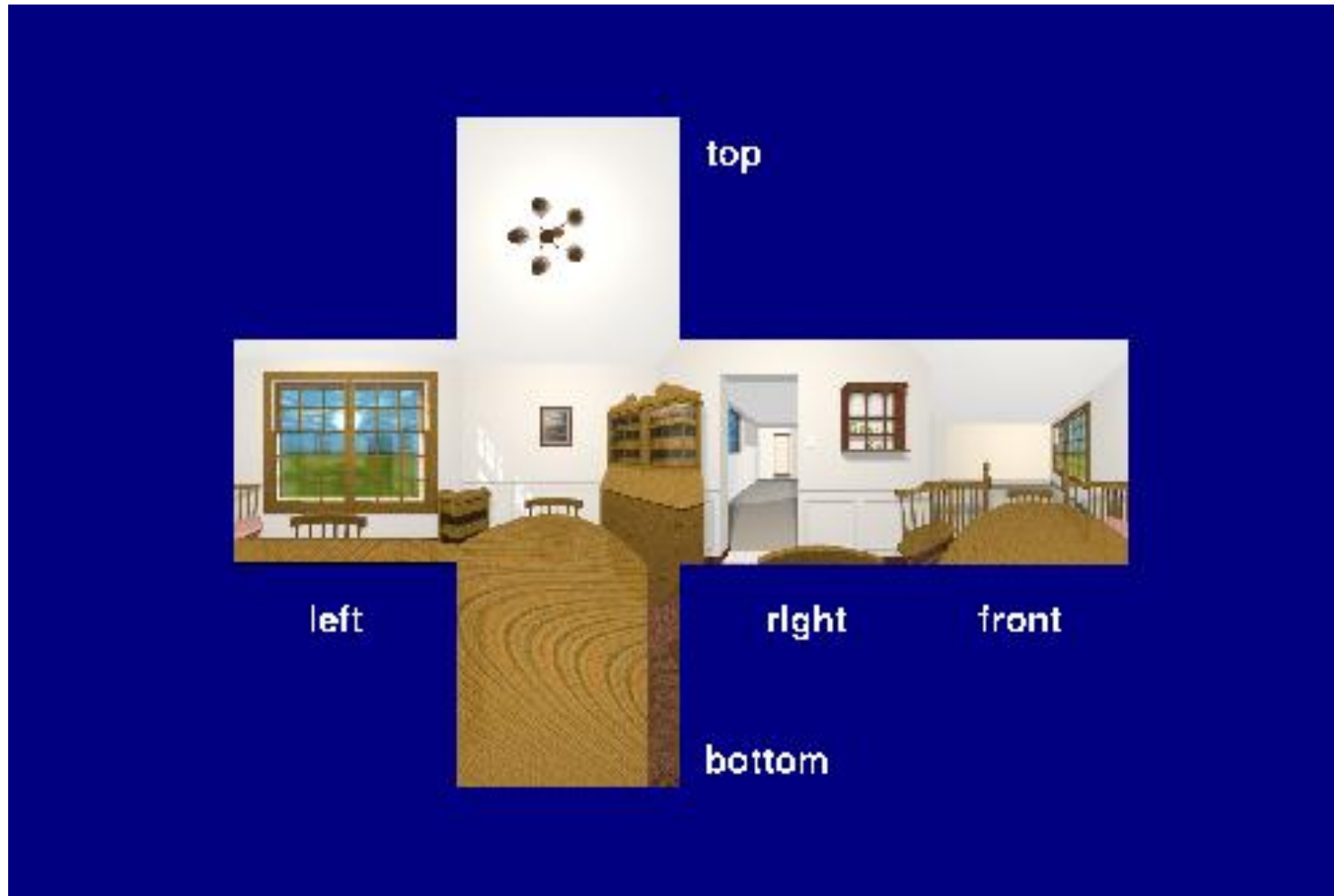


(a) View rays intersect the cube surface



(b) Mapping the environment images onto the cube faces.

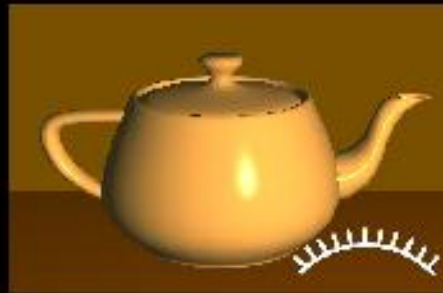
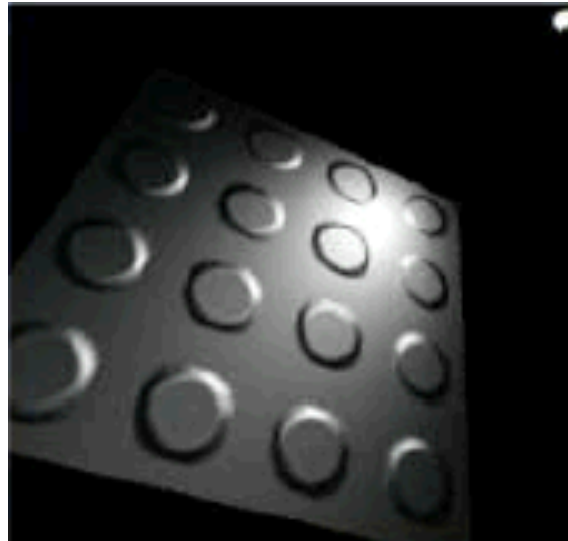
Env mapping by a cube - example



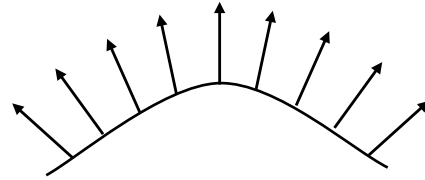
Bump mapping

- ❑ Developed in 1978 by J.F. Blinn
- ❑ Very efficient and simple method to present deformed surfaces (e.g.)
- ❑ Method:
 1. Angular modification of the normal vector according with a 2D model of deformation
 2. Apply the local reflection model to compute the light intensity
- ❑ Requirements:
 - Vector modification should be independent of object orientation and position
 - The texture should change (animate) as the object changes
 - The normal vector deformation should be defined in local coordinates relative to the surface point

Bump mapping - examples



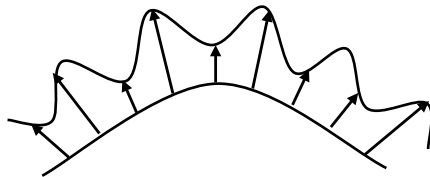
Bump mapping - phases



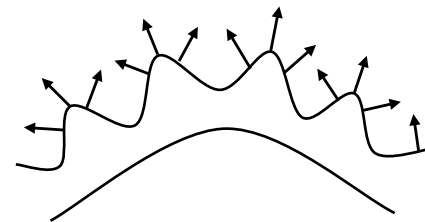
$O(u)$ Initial surface



$B(u)$ Deformation scheme



$O'(u)$ - Modification of $O(u)$
by $B(u)$



$N'(u)$ - Normal vectors of
the deformed surface

Bump mapping - definitions

- $O(u, v)$ – parametrical definition of the object surface. O is a point on the surface.
- Normal vector is: $N = O_u \times O_v$
- The deformation function is defined relative to two vectors A and B , tangent at the surface:

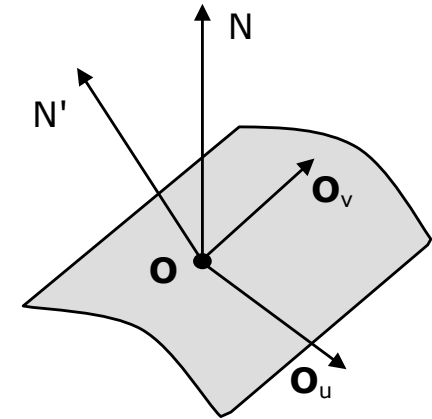
$$A = N \times O_v \quad B = N \times O_u$$

- Deformation function modifies the normal N to N' :

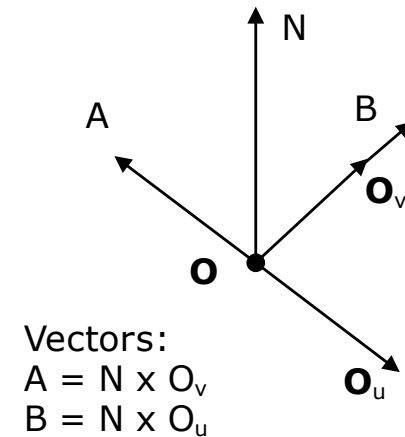
$$D = B_u A - B_v B$$

$$N' = N + D$$

Where B_u and B_v are the partial derivatives of the perturbation function $B(u, v)$

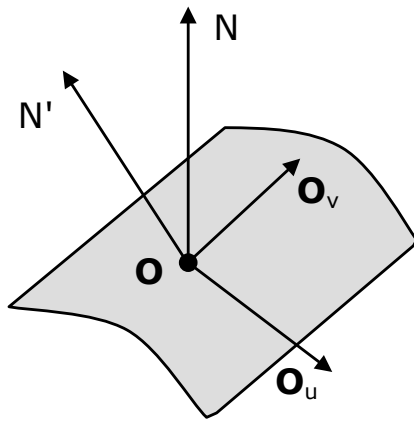


Normal vector at O is
 $N = O_u \times O_v$

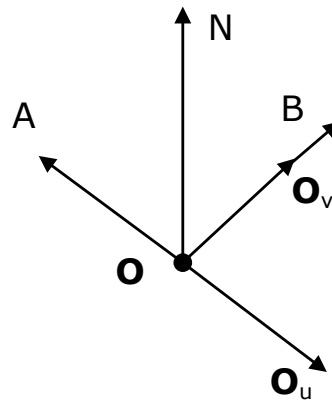


Vectors:
 $A = N \times O_v$
 $B = N \times O_u$

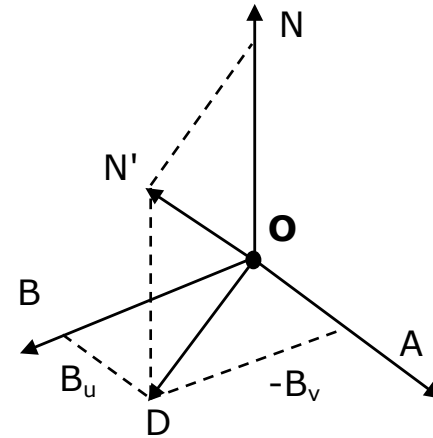
Bump mapping - vectors



Normal vector at O is
 $N = O_u \times O_v$



Vectors:
 $A = N \times O_v$
 $B = N \times O_u$



Vectors:
 $D = B_u A - B_v B$
 $N' = N + D$

Bump mapping - arguments

- $O'(u, v)$ is the deformed surface by the perturbation function and normal vector:

$$O'(u, v) = O(u, v) + B(u, v) \frac{N}{|N|}$$

- The derivatives of the object surface at point O are:

$$O'_u = O_u + B_u \frac{N}{|N|} + B \left(\frac{N}{|N|} \right)_u$$

$$O'_v = O_v + B_v \frac{N}{|N|} + B \left(\frac{N}{|N|} \right)_v$$

- The function $B(u, v)$ is quite small regarding the spatial extension, and the last term can be neglected

$$N'(u, v) = O_u \times O_v + B_u \left(\frac{N}{|N|} \times O_v \right) + B_v \left(O_u \times \frac{N}{|N|} \right) + B_u B_v \left(\frac{N \times N}{|N|^2} \right)$$

Bump mapping - arguments

The first term is the normal vector to the surface, and the last is zero. Then:

$$D = B_u (N \times O_v) - B_v (N \times O_u)$$

where, $N/|N|$ is considered as N , and

$$D = B_u A - B_v B$$

D is a vector coplanar with A and B , having the components B_u and $-B_v$.

N' is given by:

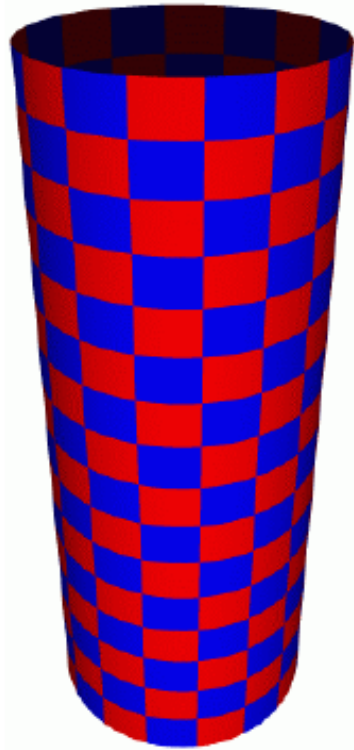
$$N' = N + D$$

That is the estimated result for the perturbed vector

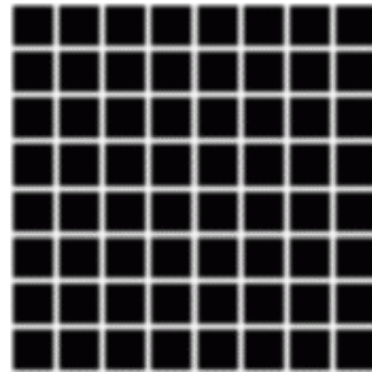
Bump mapping - issues

- ❑ Bump mapping depends on the scaling factor on the object
- ❑ For instance, if the scale factor is 2, then the N vector increases by 4, but the length of D increases by 2
- ❑ The perturbation function B is not scaled at all. Therefore, the bumps become smooth, as well as the object surface stretches out

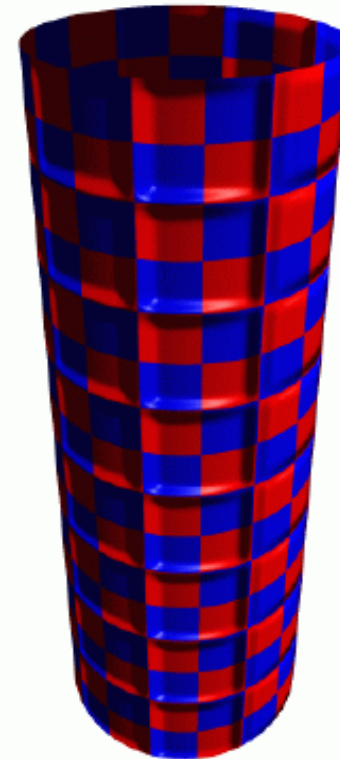
Bump mapping - examples



Cylinder w/Diffuse Texture Map

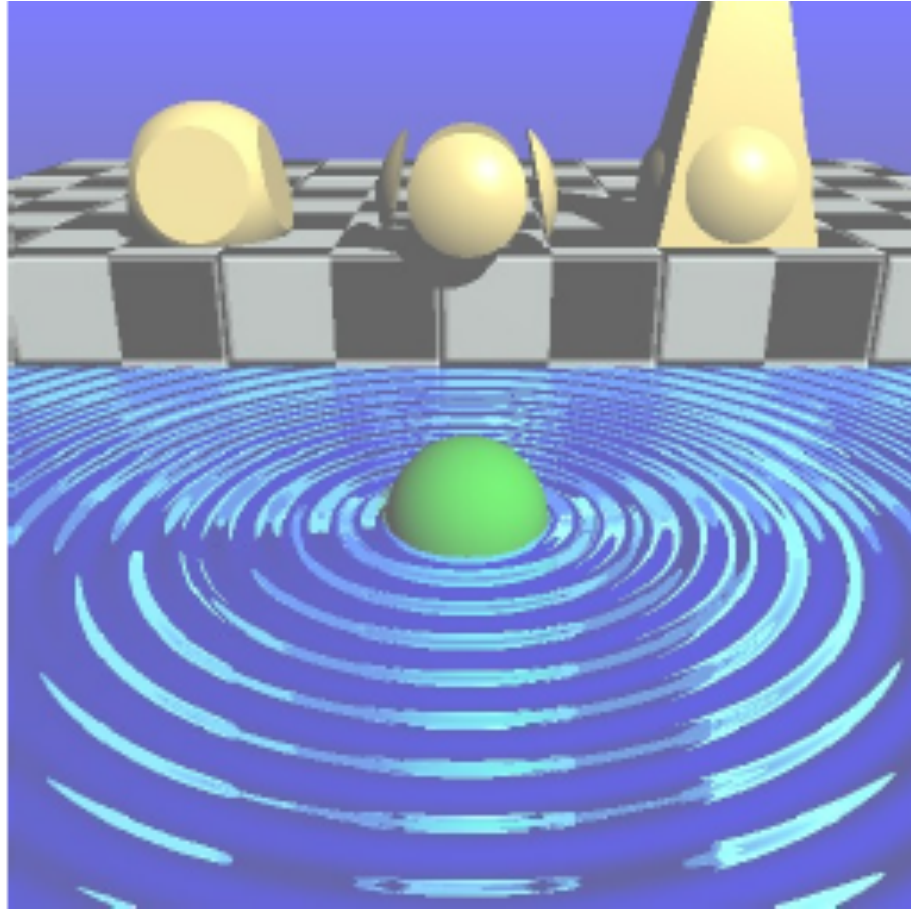


Bump Map



Cylinder w/Texture Map & Bump Map

Bump mapping - examples



Questions and problems

1. Exemplify and explain how is computed the value of a point on the 3D object by mapping of a 3D normalized texture.
2. Explain the mechanism of bump mapping to obtain the image of: (a) water surface with circular waves; (b) cylindrical bumps.

